

Nicole Ye

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TECHNICAL SKILLS

Firmware & Embedded:: ESP32, Raspberry Pi Pico, C++, Java, PlatformIO, I2C, UART,
Electronics: KiCad, EasyEDA, PCB Schematic & Layout, Soldering, Multimeter, Oscilloscope, Power Supply
Sensors & Protocols: VL53L4CX ToF, RDM630 RFID, Hall-Effect, Encoders,
Mechanical/CAD: Onshape, SolidWorks, BambuStudio, 3-D Printing, Sheet Metal, Snap-Fits

EXPERIENCE

- Prometheus Industries (GreaseTech) — Mechatronics Engineering Intern** Jan. 2026 – Apr. 2026
- Integrated **VL53L4CX** time-of-flight sensor via **I2C on ESP32** for threshold-based canister detection, replacing an unreliable optical sensor and eliminating false-positive reads in production firmware.
 - Developed hall-effect sensor firmware (C++) for rotational grease-dispense tracking, enabling real-time dispense-volume monitoring across 14+ assembled production units.
 - Resolved RFID misreads across 200+ sorted tags by characterizing per-tag-type antenna offsets and iterating nozzle grip geometry in Onshape to a unified 84 mm reference point, eliminating read failures across all tag variants.
 - Accomplished a 35% reduction in system wiring complexity into a single dual-layer PCB, by designing a custom integration board that routes an ESP32 microcontroller directly to an RDM6300 RFID reader, a Hall effect sensor, and a 12V motor-driving relay circuit.
 - Iterated protective electronics enclosure across 3 CAD revisions, reducing volume by 30% while integrating wire channels, standoffs, and heat inserts.
- Team 2198 FIRST Robotics Team — Captain/Electrical Lead** Sep. 2021 – Jun. 2025
- Captained a 30+ member team compete across **4 seasons**, achieving a **top-10 regional finish** in 2024–2025.
 - Engineered electrical system using [TKO Circuit Simulator](#), integrating colour sensors, magnetic sensors, and encoders for real-time game-piece detection **10+ electrical components**.
 - Programmed AprilTag-based field-relative localization in **Java** using Limelight vision co-processor, enabling automated robot alignment to scoring positions and reducing driver repositioning time during matches.
 - Designed 5+ sub-mechanisms (elevator, intake, climber, shooter) in **Onshape**, coordinating design-to-manufacture handoff within a **6-week** build season.

PROJECTS

- Automated RFID Conveyor Sorting System** Jan. 2026 – Apr. 2026
- Designed a dual-motor conveyor using an ESP32 and RDM630 reader to decode and sort RFID-tagged canisters.
 - Programmed a C++ deterministic state machine to sync UART RFID decoding with PWM servo diversion.
 - Classified canisters via 7- and 0- prefixed tags, hitting 720 tags/hr and 100% accuracy over 50+ runs.
- WALL-E Animatronic Robot** Jan. 2026 – Present
- Modeling an 3-D printed frame in Onshape with 6+ servo actuators and an OV2640 camera module.
 - Writing PlatformIO firmware to unify I2C sensor polling, PWM servo control, ESPNOW wireless communication.
 - Designing a custom 2-layer ESP32 PCB integrating discrete buck regulators and dedicated servo header arrays.

AWARDS

Queen's Chancellor Scholarship Receipt (\$48,000), University of Toronto Book Scholar Award
L'Amoureux C.I.'s Student of the Year, Innovation in Control Award - ONT FRC 2024 Humber College Event

EDUCATION

University of Waterloo Sep. 2025 - Apr. 2030
Honour's Electrical Engineering, Bachelor's of Applied Science *Waterloo, ON, Canada*